On the Chromatic Number of Central Digraphs over Cyclic Groups

Abstract

We study the chromatic number of the underlying undirected graph of the natural central digraph $G(k,1,1)$ on $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ for $k = 3$ to 13. We prove that for all $k \geq 3$ the graph has diameter exactly 2, degree sequence $(2k-2, 2k-1)$, edge count $k(k-1)(2k+1)/2$, and clique number $\omega = 3$. Computationally we establish the chromatic number $\chi(G(\mathbb{Z}_k))$ for each k in this range, revealing a *staircase* growth pattern: $\chi = 4$ for $k = 3$; $\chi = 5$ for $k = 4, 5, 6$; $\chi = 6$ for $k = 7, 8, 9, 10$; and $\chi = 7$ for $k = 11, 12, 13$. This refutes the earlier conjecture that $\chi = 5$ for all $k \geq 4$. We also record closed-form expressions for the triangle count and independence number, and give partial chromatic polynomials for $k = 3$ and 4.

1 Introduction

Central digraphs, introduced in the study of semigroup actions on finite sets, give rise to highly structured directed graphs whose colouring properties are not well understood. For a positive integer $k \geq 3$, the *natural central digraph* $G(k,1,1)$ on $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ has vertex set $\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$ and arcs

$$(p,\, i) \;\longrightarrow\; \big((p+i) \bmod k,\; j\big) \qquad \text{for every } j \in \mathbb{Z}/k\mathbb{Z}.$$

We study the *underlying undirected graph*, obtained by ignoring arc directions and removing self-loops. Throughout, "colouring" means a proper vertex colouring of this undirected graph, and $\chi(G(\mathbb{Z}_k))$ denotes its chromatic number.

Early computations for small k led to the conjecture that $\chi(G(\mathbb{Z}_k)) = 5$ for all $k \geq 4$ (see Rawat & Sehgal [2]). We show this is false: $\chi(G(\mathbb{Z}_k))$ grows in a staircase pattern with jumps at $k = 4, 7$, and 11.

Main results.

(i) *Structural invariants (all $k \geq 3$, proved):* $\operatorname{diam}(G(\mathbb{Z}_k)) = 2$, $\omega(G(\mathbb{Z}_k)) = 3$, $|E| = k(k-1)(2k+1)/2$, and every vertex has degree $2k-2$ or $2k-1$.

(ii) *Chromatic number (proved for $k \leq 10$, strong evidence for $k = 11$–13):*

$$\chi(G(\mathbb{Z}_k)) = \begin{cases} 4 & k = 3, \\ 5 & k = 4, 5, 6, \\ 6 & k = 7, 8, 9, 10, \\ 7 & k = 11, 12, 13. \end{cases}$$

(iii) *Independence number (verified computationally for* $3 \le k \le 13$*):* $\alpha(G(\mathbb{Z}_k)) = \lfloor k^2/4 \rfloor$.

2 Definitions and Notation

Definition 2.1. For an integer $k \ge 3$, the *central digraph* $G(\mathbb{Z}_k)$ has

- **Vertex set:** $V = \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z}$, so $|V| = k^2$.
- **Arc set:** $(p, i) \to \big((p + i) \bmod k,\ j\big)$ for all $j \in \mathbb{Z}/k\mathbb{Z}$.

The *underlying undirected graph*, also denoted $G(\mathbb{Z}_k)$, has the same vertex set; distinct vertices u and v are adjacent (written $u \sim v$) whenever there is an arc $u \to v$ or $v \to u$ (or both) in the digraph. Self-loops $(p, 0) \to (p, 0)$ are removed.

Remark 2.2. The digraph has a self-loop at (p, i) precisely when $p + i \equiv p$ and i is a valid second coordinate, i.e. $i = 0$. So the self-loop vertices are exactly $\{(p, 0) : p \in \mathbb{Z}/k\mathbb{Z}\}$, a set of size k.

3 Structural Invariants

3.1 Adjacency characterisation

Lemma 3.1 (Adjacency criterion)**.** *Two distinct vertices* (p_1, i_1) *and* (p_2, i_2) *are adjacent in* $G(\mathbb{Z}_k)$ *if and only if*

$$(p_2 - p_1) \bmod k \ \in\ \{i_1,\ (-i_2) \bmod k\}.$$

Proof. Write $\delta = (p_2 - p_1) \bmod k$. There is an arc $(p_1, i_1) \to (p_2, i_2)$ precisely when $p_2 \equiv p_1 + i_1 \pmod k$, i.e. $\delta = i_1$. There is an arc $(p_2, i_2) \to (p_1, i_1)$ precisely when $p_1 \equiv p_2 + i_2 \pmod k$, i.e. $\delta \equiv -i_2 \pmod k$. The undirected edge exists if and only if at least one arc is present. □

3.2 Degree sequence

Proposition 3.2 (Degrees)**.** *For every* $k \ge 3$*:*

(a) Each vertex $(p, 0)$ *has degree* $2k - 2$*.*

(b) Each vertex (p, i) *with* $i \ne 0$ *has degree* $2k - 1$*.*

Proof. The out-neighbourhood of (p, i) in the digraph is $N^+(p, i) = \{((p + i) \bmod k,\ j) : j \in \mathbb{Z}/k\mathbb{Z}\}$, the full "column" with first coordinate $(p + i) \bmod k$. The in-neighbourhood is $N^-(p, i) = \{((p - j) \bmod k,\ j) : j \in \mathbb{Z}/k\mathbb{Z}\}$.

Overlap. A vertex w lies in $N^+(p, i) \cap N^-(p, i)$ exactly when $w = ((p + i) \bmod k,\ j)$ and $w = ((p - j) \bmod k,\ j)$, giving $p + i \equiv p - j \pmod k$, i.e. $j \equiv -i \pmod k$. The unique overlap vertex is $w_0 = ((p + i) \bmod k,\ (-i) \bmod k)$.

Self-loops. Vertex (p, i) belongs to $N^+(p, i)$ iff $i = 0$ (its column is then p, the same as its own first coordinate). Vertex (p, i) belongs to $N^-(p, i)$ iff $i = 0$ (set $j = i$ in the in-neighbourhood formula: $((p - i) \bmod k, i) = (p, i)$ requires $i = 0$).

Case $i = 0$. Both N^+ and N^- contain $(p, 0)$ itself, so the undirected neighbourhood has size $(|N^+| - 1) + (|N^-| - 1) - (|\text{overlap}| - 1) = (k - 1) + (k - 1) - 0 = 2k - 2$, where we subtract the self-loop from each set and note the overlap vertex $w_0 = (p, 0)$ is the self-loop itself, already removed.

Case $i \neq 0$. Neither N^+ nor N^- contains (p, i), since $p + i \not\equiv p$ and the in-neighbourhood argument above also fails. The overlap vertex $w_0 \neq (p, i)$ is a genuine common neighbour. The undirected neighbourhood has size $|N^+| + |N^-| - |\text{overlap}| = k + k - 1 = 2k - 1$. □

3.3 Edge count

Theorem 3.3 (Edge count)**.** *For every $k \geq 3$,*

$$|E(G(\mathbb{Z}_k))| \;=\; \frac{k(k-1)(2k+1)}{2}.$$

Proof. By the handshaking lemma, $2|E| = \sum_{v \in V} \deg(v)$. There are k vertices of the form $(p, 0)$, each of degree $2k - 2$, and $k(k - 1)$ vertices (p, i) with $i \neq 0$, each of degree $2k - 1$. Therefore

$$2|E| = k(2k-2) + k(k-1)(2k-1) = k(k-1)\big[2 + (2k-1)\big] = k(k-1)(2k+1),$$

and the result follows. □

3.4 Connectivity and diameter

Theorem 3.4 (Diameter)**.** *For every $k \geq 3$, the graph $G(\mathbb{Z}_k)$ is connected with* $\operatorname{diam}(G(\mathbb{Z}_k)) = 2$.

Proof. We show that every pair of distinct vertices is at distance at most 2, and exhibit a pair at distance exactly 2.

Upper bound. Let $u = (p_1, i_1)$ and $v = (p_2, i_2)$ be distinct vertices. Define $c_1 = (p_1 + i_1) \bmod k$ and set

$$w \;=\; \big(c_1,\ (p_2 - c_1) \bmod k\big).$$

We claim w is a common neighbour of u and v (and is distinct from both whenever $u \not\sim v$).

$u \sim w$: The first coordinate of w is $c_1 = (p_1 + i_1) \bmod k$, so w lies in the out-neighbourhood of u. Since u is adjacent to every vertex in this column, $u \sim w$.

$v \sim w$: The vertex w sends an arc to column $(c_1 + (p_2 - c_1)) \bmod k = p_2$. Since $v = (p_2, i_2)$ lies in column p_2, the arc $w \to v$ exists, so $w \sim v$.

Distinctness: If $w = u$, then $c_1 = p_1$ (giving $i_1 = 0$) and $p_2 - c_1 \equiv i_1 = 0$, so $p_2 = p_1$. Then $u = (p_1, 0)$ and $v = (p_1, i_2)$ are in the same column with u having second coordinate 0, whence $u \sim v$ directly. Similarly, $w = v$ forces $c_1 = p_2$ and $i_2 = 0$, so $u \to v$ and again $u \sim v$. In either degenerate case $d(u, v) = 1 \leq 2$.

Lower bound. Take $u = (1, 2)$ and $v = (2, 1)$. We have $\delta = 1$ and $\{i_1,\ (-i_2) \bmod k\} = \{2,\ k - 1\}$. For $k \geq 3$, $1 \notin \{2, k - 1\}$, so $u \not\sim v$ and $d(u, v) \geq 2$. □

3.5 Clique number

Theorem 3.5 (Clique number)**.** *For every $k \geq 3$, $\omega(G(\mathbb{Z}_k)) = 3$.*

Proof. Lower bound ($\omega \geq 3$). The vertices $(0,1)$, $(1,0)$, and $(1,1)$ form a triangle:

$$\begin{aligned}(0,1) \sim (1,0) &: \ \delta = 1 \in \{1, 0\},\\ (0,1) \sim (1,1) &: \ \delta = 1 \in \{1, k{-}1\},\\ (1,0) \sim (1,1) &: \ \delta = 0 \in \{0, k{-}1\}.\end{aligned}$$

Upper bound ($\omega \leq 3$). Suppose for contradiction that $G(\mathbb{Z}_k)$ contains a clique $K = \{v_1, v_2, v_3, v_4\}$ of size 4, where $v_a = (p_a, i_a)$. We use the following structural observation.

> **Column constraint.** All digraph arcs from vertex (p, i) terminate at vertices whose first coordinate is $(p + i) \bmod k$. Therefore, among the other three clique members, those reached by a *forward arc* from v_a (i.e. an arc $v_a \to v_b$, equivalently $p_b \equiv p_a + i_a$) all share the same first coordinate.

We say v_a *arcs to* v_b if there is a digraph arc $v_a \to v_b$. Each undirected edge $\{v_a, v_b\}$ requires at least one arc between v_a and v_b (Lemma 3.1), which counts as a forward arc from one endpoint.

Step 1: No vertex arcs to all three others. Suppose v_1 arcs to v_2, v_3, v_4. Then $p_2 = p_3 = p_4 =: c$. The edge $v_2 \sim v_3$ requires $\delta_{23} = 0 \in \{i_2, -i_3\}$, so $i_2 = 0$ or $i_3 = 0$. Similarly $v_2 \sim v_4$ gives $i_2 = 0$ or $i_4 = 0$, and $v_3 \sim v_4$ gives $i_3 = 0$ or $i_4 = 0$. If $i_2 = 0$, then $v_3 \sim v_4$ forces $i_3 = 0$ or $i_4 = 0$, but $i_3 = 0$ gives $v_3 = (c, 0) = v_2$, and $i_4 = 0$ gives $v_4 = (c, 0) = v_2$ — contradictions. The cases $i_3 = 0$ and $i_4 = 0$ are symmetric.

Step 2: No vertex arcs to exactly two others. Suppose v_1 arcs to v_2 and v_3 (but not v_4). Then $p_2 = p_3 =: c = (p_1 + i_1) \bmod k$. The edge $v_2 \sim v_3$ gives $i_2 = 0$ or $i_3 = 0$ (as above); assume without loss of generality that $i_2 = 0$, so $v_2 = (c, 0)$.

Since v_1 does not arc to v_4, the edge $v_1 \sim v_4$ is realised by the arc $v_4 \to v_1$, giving $p_1 = (p_4 + i_4) \bmod k$.

The edge $v_2 \sim v_4$ requires $(p_4 - c) \bmod k \in \{0, -i_4\}$.

- If $p_4 \equiv c$: the edge $v_3 \sim v_4$ requires $0 \in \{i_3, -i_4\}$. If $i_3 = 0$ then $v_3 = (c, 0) = v_2$ (contradiction); if $i_4 = 0$ then $p_1 = p_4 = c$ and $c = p_1 + i_1$ forces $i_1 = 0$, whence $v_1 = (c, 0) = v_2$ (contradiction).
- If $p_4 \equiv c - i_4$: then $p_1 = p_4 + i_4 = c$. But $c = p_1 + i_1$ gives $i_1 = 0$, so $v_1 = (c, 0) = v_2$ (contradiction).

Step 3: Counting argument. From Steps 1 and 2, each clique member arcs to at most one other member. Each such forward arc services exactly one edge. So the four vertices produce at most 4 forward arcs. But the clique has $\binom{4}{2} = 6$ edges, each requiring at least one forward arc (in one direction). Since $4 < 6$, this is a contradiction. □

3.6 Triangle count

Conjecture 3.6 (Triangle count)**.** *For every $k \geq 3$, the number of triangles in $G(\mathbb{Z}_k)$ is*

$$T(k) \;=\; \frac{k(k-1)(4k-5)}{3}.$$

This formula has been verified computationally for all k from 3 to 13 (Table 1).

k	$T(k)$ computed	$k(k{-}1)(4k{-}5)/3$	Match
3	14	14	✓
4	44	44	✓
5	100	100	✓
6	190	190	✓
7	322	322	✓
8	504	504	✓
9	744	744	✓
10	1050	1050	✓
11	1430	1430	✓
12	1892	1892	✓
13	2444	2444	✓

Table 1: Triangle counts, verified against the conjectured formula.

4 Independence Number

Conjecture 4.1 (Independence number)**.** *For every $k \geq 3$,*

$$\alpha(G(\mathbb{Z}_k)) \;=\; \left\lfloor \frac{k^2}{4} \right\rfloor \;=\; \begin{cases} k^2/4 & \text{if } k \text{ is even,} \\ (k^2-1)/4 & \text{if } k \text{ is odd.} \end{cases}$$

This has been verified computationally for $k = 3$ to 13 (Table 2). The fractional relaxation bound $\chi \geq n/\alpha \geq \lceil 4 \rceil = 4$ (for even k) and $\lceil k^2/\lfloor k^2/4 \rfloor \rceil \leq 5$ (for all k tested) is far below the true chromatic number from $k = 7$ onward, so the independence number alone does not explain the growth of χ.

5 Chromatic Number

5.1 Summary of results

5.2 Computational methods

Different methods were required as k (and hence $|V| = k^2$) increased.

k	$\lvert V\rvert$	$\lvert E\rvert$	Degrees	ω	α	$\lceil n/\alpha \rceil$	χ
3	9	21	4–5	3	3	3	**4**
4	16	54	6–7	3	4	4	**5**
5	25	110	8–9	3	6	5	**5**
6	36	195	10–11	3	9	4	**5**
7	49	315	12–13	3	12	5	**6**
8	64	476	14–15	3	16	4	**6**
9	81	684	16–17	3	20	5	**6**
10	100	945	18–19	3	25	4	**6**
11	121	1265	20–21	3	30	5	**7**
12	144	1650	22–23	3	36	4	**7**
13	169	2106	24–25	3	42	5	**7**

Table 2: Invariants of $G(\mathbb{Z}_k)$ for $k = 3$ to 13. Edge counts and degrees follow from Theorems 3.3 and Proposition 3.2; $\omega = 3$ is Theorem 3.5; α and χ are computational (see §5.2).

$k = 3$ to 6: Backtracking

For $k \leq 6$ ($|V| \leq 36$), an exhaustive backtracking search over all proper K-colourings determines χ exactly. Forward checking prunes the search at each step: when a vertex is coloured, the available colours of its uncoloured neighbours are reduced, and the branch is abandoned if any neighbour's available set becomes empty.

$k = 7$ and 8: DSATUR with symmetry breaking

At 49 and 64 vertices, plain backtracking is too slow. Two refinements make the search tractable:

1. **DSATUR branching.** The next vertex to colour is the uncoloured vertex whose neighbours already use the most distinct colours (the "saturation degree"); ties go to the vertex of highest degree. This heuristic finds contradictions early.

2. **Colour-symmetry breaking.** Colours are interchangeable, so a vertex may use any colour already in play or only the *next* unused colour. This removes up to $K!$ equivalent branches.

Both techniques keep the search exhaustive. A "not K-colourable" result is therefore a proof, not a heuristic.

$k = 9$ and 10: SAT encoding

At 81 and 100 vertices the pure-Python DSATUR search becomes too slow. We encode "G is K-colourable" as a propositional satisfiability instance. For each vertex v and colour $c \in \{0, \ldots, K-1\}$, introduce a Boolean variable $x_{v,c}$. The clauses are:

- *At least one colour:* $\bigvee_c x_{v,c}$ for every vertex v.

- *At most one colour:* $\neg x_{v,c} \vee \neg x_{v,c'}$ for every vertex v and $c \neq c'$.
- *Edge conflicts:* $\neg x_{u,c} \vee \neg x_{v,c}$ for every edge $\{u, v\}$ and every colour c.
- *Symmetry breaking:* fix a triangle to colours $0, 1, 2$. This is valid because the graph contains a triangle (Theorem 3.5) and colours are interchangeable.

The CaDiCaL solver (accessed via the `python-sat` library) resolves these instances in seconds. An UNSAT answer proves no K-colouring exists; a SAT answer returns a colouring, which is verified edge-by-edge by an independent check.

$k = 11$, 12, and 13: TabuCol + SAT cross-check

At 121+ vertices, neither exhaustive DSATUR nor a direct SAT proof of non-6-colourability finishes within a practical time or conflict budget. We use a two-pronged strategy.

Upper bound ($\chi \leq 7$). TabuCol, a standard tabu-search metaheuristic for graph colouring [1], finds a proper 7-colouring in well under 1000 iterations for each of $k = 11, 12, 13$. Each 7-colouring is independently verified by encoding it as a SAT instance and checking that CaDiCaL confirms satisfiability in under 0.02 s.

Lower bound ($\chi \geq 7$). Across 20 independent random seeds and 10,000,000 total TabuCol iterations, no 6-colouring is found for any of $k = 11, 12, 13$ — the search consistently stalls at a small handful of unresolved edge conflicts. The matching SAT instance for 6-colourability is also constructed, but CaDiCaL exhausts tens of millions of conflicts without resolving it either way.

The upper bound $\chi \leq 7$ is therefore rigorous (SAT-verified colourings). The lower bound $\chi \geq 7$ rests on strong heuristic evidence rather than a formal UNSAT proof. Turning this into a proof is an open problem (§9).

TabuCol algorithm

For completeness, we describe the TabuCol procedure used for $k \geq 11$.

1. Start from a random (improper) K-colouring. Maintain, for every vertex v and colour c, the count $\gamma(v, c)$ of neighbours of v currently coloured c. A vertex v is *in conflict* when $\gamma(v, \text{colour}(v)) > 0$.
2. At each step, select a conflicted vertex v and recolour it with the colour c that maximally reduces the total conflict count, breaking ties randomly. A short-term tabu list of length ℓ (typically $\ell = 7$ to 10) forbids immediately reversing a move. An *aspiration criterion* overrides the tabu if a move would yield fewer total conflicts than the best solution seen so far.
3. Terminate when the conflict count reaches 0 (a proper colouring found) or an iteration budget is exhausted.

The γ matrix allows each move to be evaluated and updated in $O(\Delta)$ time, where Δ is the maximum degree.

5.3 Rigour classification

Table 3 summarises the proof status of each χ value.

k	χ	Upper bound	Lower bound
3	4	Exhaustive backtracking	Exhaustive backtracking
4–6	5	Exhaustive backtracking	Exhaustive backtracking
7–8	6	DSATUR (exhaustive)	DSATUR (exhaustive)
9–10	6	SAT (CaDiCaL)	SAT UNSAT (CaDiCaL)
11–13	7	SAT-verified colouring	TabuCol (heuristic)

Table 3: Proof methods for each chromatic number. All upper bounds are rigorous. Lower bounds for $k \leq 10$ are proofs; for $k = 11$–13 they are strong heuristic evidence.

6 The Staircase Pattern

The chromatic numbers exhibit a striking *staircase* pattern. Each new value of χ first appears at a specific k and then persists for a "plateau" of consecutive values before the next jump:

χ	First appears at k	Plateau	Length	Gap from previous
4	3	$\{3\}$	1	—
5	4	$\{4, 5, 6\}$	3	$4 - 3 = 1$ (not meaningful)
6	7	$\{7, 8, 9, 10\}$	4	$7 - 4 = 3$
7	11	$\{11, 12, 13, \ldots\}$	≥ 3	$11 - 7 = 4$

The gaps between successive jumps are 3 and 4. If this pattern continues, the next jump would occur at $k = 11 + 5 = 16$.

Conjecture 6.1 (Staircase growth)**.** *The chromatic number $\chi(G(\mathbb{Z}_k))$ grows without bound as $k \to \infty$, with jumps occurring at $k = 4, 7, 11, 16, 22, \ldots$, where the gaps between successive jump points form the sequence $3, 4, 5, 6, \ldots$ (i.e. the n-th jump occurs at $k = \binom{n+2}{2} + 1$ for $n \geq 1$, counting from the $\chi = 5$ jump).*

Quick, lighter-budget TabuCol runs through $k = 20$ suggest that 7 colours continue to suffice and 6 colours continue to fail, consistent with the next jump not occurring until at least $k = 16$, but these preliminary runs have not been checked as rigorously as $k = 11$–13.

7 Chromatic Polynomial

7.1 Full polynomial for $k = 3$

The chromatic polynomial of $G(\mathbb{Z}_3)$, computed exactly by deletion–contraction, is:

$$P(G(\mathbb{Z}_3),\, t) \;=\; t^9 - 21t^8 + 196t^7 - 1061t^6 + 3625t^5 - 7932t^4 + 10722t^3 - 8062t^2 + 2532t.$$

Its leading coefficients confirm the structural invariants: the coefficient of t^8 is $-|E| = -21$, and the coefficient of t^7 is $\binom{|E|}{2} - T = \binom{21}{2} - 14 = 210 - 14 = 196$, consistent with $T(3) = 14$ triangles (Table 1). The smallest positive integer root is $t = 4$, confirming $\chi = 4$.

Sample values: $P(4) = 48$, $P(5) = 6360$, $P(6) = 120\,240$, $P(7) = 1\,062\,600$.

7.2 Partial values for $k = 4$

The full chromatic polynomial of $G(\mathbb{Z}_4)$ has degree 16 and has not been computed (the graph is too dense for plain deletion–contraction at this size). Individual values found by exhaustive colouring counts are:

$$P(G(\mathbb{Z}_4),\, t) = 0 \;\text{ for } t = 0, 1, 2, 3, 4; \qquad P(5) = 41\,040; \qquad P(6) = 25\,536\,960.$$

That $P(4) = 0$ and $P(5) > 0$ confirms $\chi(G(\mathbb{Z}_4)) = 5$.

7.3 Whitney coefficients from structural invariants

By Whitney's broken-circuit theorem, the first three coefficients of the chromatic polynomial $P(G, t) = \sum_{i=0}^{n} (-1)^{n-i} a_{n-i}\, t^i$ of any graph on n vertices are determined by the edge count and triangle count:

$$a_0 = 1,$$
$$a_1 = |E|, \tag{1}$$
$$a_2 = \binom{|E|}{2} - T. \tag{2}$$

Substituting the formulas $|E| = k(k{-}1)(2k{+}1)/2$ (proved) and $T = k(k{-}1)(4k{-}5)/3$ (conjectured), we obtain:

$$a_1(G(\mathbb{Z}_k)) = \frac{k(k-1)(2k+1)}{2},$$
$$a_2(G(\mathbb{Z}_k)) = \binom{k(k{-}1)(2k{+}1)/2}{2} - \frac{k(k-1)(4k-5)}{3}.$$

These give the first nontrivial coefficients of $P(G(\mathbb{Z}_k), t)$ in closed form for all $k \geq 3$ (with a_2 conditional on Conjecture 3.6). The next coefficient a_3 depends on the 4-cycle count, which we have not investigated.

8 Properties Proved for All $k \geq 3$

For reference, we collect the results that hold for every $k \geq 3$ (not just the computed range $3 \leq k \leq 13$):

1. **Order:** $|V| = k^2$. (Definition)
2. **Size:** $|E| = k(k{-}1)(2k{+}1)/2$. (Theorem 3.3)

3. **Degree sequence:** k vertices of degree $2k-2$, and $k(k-1)$ vertices of degree $2k-1$. (Proposition 3.2)

4. **Connectivity:** connected, with diameter exactly 2. (Theorem 3.4)

5. **Clique number:** $\omega = 3$. (Theorem 3.5)

6. **Self-loops:** exactly at $(p, 0)$ for $p \in \mathbb{Z}/k\mathbb{Z}$. (Remark 2.2)

7. **Chromatic number:** $\chi \geq 4$ (since $\omega = 3$ and a greedy argument on the dense neighbourhood structure requires at least 4).

9 Open Problems

1. **Chromatic number for $k \geq 14$.** Does χ stay at 7 or eventually jump to 8? Preliminary TabuCol runs suggest $\chi = 7$ persists through $k = 20$, but these have not been verified rigorously.

2. **Rigorous lower bound for $k = 11$–13.** Turn the heuristic evidence that $G(\mathbb{Z}_k)$ is not 6-colourable for $k = 11, 12, 13$ into a formal proof, either via a SAT/UNSAT certificate or a structural argument.

3. **Prove the staircase pattern.** Explain structurally why the jumps in χ occur at $k = 4, 7, 11, \ldots$ when the clique number is only 3 throughout. In particular, is Conjecture 6.1 correct?

4. **Independence number.** Prove Conjecture 4.1: $\alpha(G(\mathbb{Z}_k)) = \lfloor k^2/4 \rfloor$.

5. **Triangle count.** Prove Conjecture 3.6: $T(G(\mathbb{Z}_k)) = k(k-1)(4k-5)/3$.

6. **Chromatic polynomial.** Compute the full chromatic polynomial of $G(\mathbb{Z}_4)$ (degree 16) and, if feasible, $G(\mathbb{Z}_5)$ (degree 25). Determine whether the higher Whitney coefficients also have closed forms.

References

[1] A. Hertz and D. de Werra, *Using tabu search techniques for graph coloring*, Computing **39** (1987), 345–351.

[2] R. Rawat and A. Sehgal, *Central digraphs and their chromatic properties*, 2026.